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embedplyr: Tools for Working With Text Embeddings

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Software

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Statement of Need

Dense vector embeddings are the fundamental building block of modern natural language processing (Lauriola, Lavelli, & Aiolli, 2022). The ability to represent the meaning of texts as a continuous, high dimensional space underlies the recent successes of large language models (LLMs). Embeddings have also revolutionized research methods and inspired new theoretical frameworks in linguistics, psychology, sociology, and neuroscience (see e.g. Duran, Paxton, & Fusaroli, 2019; Feuerriegel et al., 2025; Grand, Blank, Pereira, & Fedorenko, 2022; Hamilton, Leskovec, & Jurafsky, 2018; Kjell, Giorgi, & Schwartz, 2023; Kozlowski, Taddy, & Evans, 2019; Schrimpf et al., 2021). As text embeddings become ubiquitous in the social and behavioral sciences, the need for flexible, easy-to-learn tools increases. Answering this need, **embedplyr** (pronounced "embe-DEE-plier") enables common operations with word and text embeddings within a **tidyverse** (Wickham et al., 2019) and/or **quanteda** (Benoit et al., 2018) workflow, as demonstrated in Teitelbaum & Simchon (2024). **embedplyr** is designed to facilitate the use of word and text embeddings in common data manipulation and text analysis workflows, without introducing new syntax or unfamiliar data structures to R users.

Existing tools for working with embeddings in R are generally specific to particular model architectures. For example, **word2vec** (Wijffels, Watanabe, & Fomichev, 2023) provides access to word2vec models (Mikolov, Chen, Corrado, & Dean, 2013; Mikolov, Sutskever, Chen, Corrado, & Dean, 2013), **text2vec** (Selivanov, Bickel, & Wang, 2023) provides access to LDA, LSA, and GloVe models (Blei, Ng, & Jordan, 2003; Deerwester, Dumais, Furnas, Landauer, & Harshman, 1990; Pennington, Socher, & Manning, 2014), **fastText** (Mouselimis, 2024) provides access to fastText models (Bojanowski, Grave, Joulin, & Mikolov, 2017a, 2017b; Facebook, 2016; Grave, Bojanowski, Gupta, Joulin, & Mikolov, 2018), and **text** (Kjell et al., 2023) provides access to transformers-based LLMs (Wolf et al., 2020). While these tools are already invaluable in the social and behavioral sciences, using different syntax and separate data structures for each model architecture can be cumbersome. This is especially true given the prevalence of studies that make use of multiple model architectures in parallel analyses (e.g. Carrella et al., 2023; Hussain, Mata, Newell, & Wulff, 2024; Markus, Levi, Sheafer, & Shenhav, 2024). **embedplyr** fills this gap with a model agnostic approach; it can be used to work with embeddings from any model framework using syntax that will be familiar to any **tidyverse** user. While some existing tools focus on convenience functions for encouraging expert-recommended best practices (e.g. Kjell et al., 2023), **embedplyr** prioritizes flexibility—much like its namesake, **dplyr** (Wickham, François, Henry, Müller, & Vaughan, 2023). This approach yields simple, modular functions that are useful both for educating students (see Teitelbaum & Simchon, 2024) and experimenting with novel methods.

Features

Loading Pretrained Token Embeddings

embedplyr does not include tools for training new embedding models, but it can load embeddings from a file or download them from online. This is especially useful for pretrained word embedding

models like GloVe (Pennington et al., 2014), word2vec (Mikolov, Chen, et al., 2013; Mikolov, Sutskever, et al., 2013), HistWords (Hamilton et al., 2018), and fastText (Bojanowski et al., 2017b). Hundreds of these models can be conveniently downloaded from online sources with `load_embeddings()`, forgoing the need to search for model files online and juggle incompatible file types.

One particularly useful feature of `load_embeddings()` is the optional `words` parameter, which allows the user to specify a subset of words to load from the model. This allows users to work with large models, which are often too large to load into an interactive environment in their entirety.

```
# load 25d GloVe model trained on Twitter
glove_twitter_25d <- load_embeddings("glove.twitter.27B.25d")

# load words from model trained on Google Books English Fiction 1800-1810
eng.fiction.all_sgns.1800 <- load_embeddings(
    "eng.fiction.all_sgns.1800",
    words = c("word", "token", "lemma")
    )
```

The output of `load_embeddings()` is an embeddings object. An embeddings object is simply a numeric matrix with fast hash table indexing by rownames (generally tokens). This means that it can be easily coerced to a dataframe or tibble, while also allowing special embeddings-specific methods and functions, such as `predict.embeddings()` and `find_nearest()`:

```
moral_embeddings <- predict(glove_twitter_25d, c("good", "bad"))
moral_embeddings
```

```
## # 25-dimensional embeddings with 2 rows

##      dim_1 dim_2 dim_3 dim_4 dim_5 dim_6 dim_7 dim_8 dim_9 dim..
## good -0.54  0.60 -0.15 -0.02 -0.14  0.60  2.19  0.21 -0.52 -0.23 ...
## bad   0.41  0.02  0.06 -0.01  0.27  0.71  1.64 -0.11 -0.26  0.11 ...
```

```
find_nearest(glove_twitter_25d, "dog", 5L, method = "cosine")
```

```
## # 25-dimensional embeddings with 5 rows

##       dim_1 dim_2 dim_3 dim_4 dim_5 dim_6 dim_7 dim_8 dim_9 dim..
## dog    -1.24 -0.36  0.57  0.37  0.60 -0.19  1.27 -0.37  0.09  0.40 ...
## cat    -0.96 -0.61  0.67  0.35  0.41 -0.21  1.38  0.13  0.32  0.66 ...
## dogs   -0.63 -0.11  0.22  0.27  0.28  0.13  1.44 -1.18 -0.26  0.60 ...
## horse  -0.76 -0.63  0.43  0.04  0.25 -0.18  1.08 -0.94  0.30  0.07 ...
## monkey -0.96 -0.38  0.49  0.66  0.21 -0.09  1.28 -0.11  0.27  0.42 ...
```

Whereas indexing a regular matrix by rownames gets slower as the number of rows increases, **embedplyr**'s hash table indexing means that token embeddings can be retrieved in milliseconds even from models with millions of rows.

Embed Texts of Interest

Given a tidy dataframe of texts, `embed_docs()` will generate embeddings by averaging the embeddings of tokens in each text (see Ethayarajh, Duvenaud, & Hirst, 2019; Teitelbaum & Simchon, 2024, chap. 18). By default, `embed_docs()` uses a simple unweighted mean, but many other averaging methods are available.

The following example embeds three texts, which for the sake of the example will can considered to have been written by one participant diagnosed with depression, one diagnosed with anxiety, and one control.

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```r
library(dplyr)

psych_df <- tribble(
    ~id,           ~text,
    "control",     "yesterday I took my dog for a walk",
    "depression",  "I slept all day and cried in the evening",
    "anxiety",     "I kept thinking of all the things I needed to do"
    )

# add embeddings to data frame
psych_embeddings_df <- psych_df |>
    embed_docs("text", glove_twitter_25d, id_col = "id", .keep_all = TRUE)
```

`embed_tokens()` is similar to `embed_docs()`, but returns the embedding of each individual token, rather than averaging within documents.

Embed Dictionaries

Distributed Dictionary Representation (DDR) enables the application of validated psychometric lexicons (e.g. Boyd, Ashokkumar, Seraj, & Pennebaker, 2022) to rich, embedding-based semantic representation (Garten et al., 2018). This is achieved by retrieving pretrained word embeddings for each word in the dictionary, and averaging them to create a single vector—the DDR. The dictionary construct can then be measured by comparing text embeddings to the DDR.

In the following example, DDRs are constructed for high and low anxiety using example dictionaries. Once embeddings are produced for each word in the dictionary, they are averaged using `average_embedding()`. By default this is a simple mean, but `average_embedding()` also supports the geometric median (Cardot, 2022), weighted geometric median, and weighted mean, including weighting by word frequency or smooth inverse frequency (Arora, Liang, & Ma, 2017).

```r
# positive and negative construct dictionaries
high_anx_dict <- c("anxious", "overwhelmed", "nervous", "stressed")
low_anx_dict <- c("relaxed", "calm", "mellow")

# embed dictionaries
high_anx_dict_embeddings <- predict(glove_twitter_25d, high_anx_dict)
low_anx_dict_embeddings <- predict(glove_twitter_25d, low_anx_dict)

# average embeddings to create DDR
high_anx_DDR <- average_embedding(high_anx_dict_embeddings)
low_anx_DDR <- average_embedding(low_anx_dict_embeddings)
```

`average_embedding()` could be used in a similar manner to construct contextualized construct representations (Atari, Omrani, & Dehghani, 2023).

Calculate Similarity Metrics

To complete the DDR analysis initiated above, the embeddings of each the corpus texts are compared to that of the DDR. This could be done by computing cosine similarity between each text and `high_anx_DDR` (`"cosine"` is the default method for `get_sims()`). In this case however, an anchored vector is used to quantify the extent to which these texts reflect high anxiety *as opposed to low anxiety*. `method = "anchored"` gives the position of each embedding on the spectrum between two anchor points, where vectors aligned with `pos` are given a score of 1 and those aligned with `neg` are given a score of 0. This approach is also known as semantic projection (Grand et al., 2022).

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```r
anxiety_scores_df <- psych_embeddings_df |>
  get_sims(
    dim_1:dim_25,
    list(anxiety = list(pos = high_anx_DDR, neg = low_anx_DDR)),
    method = "anchored"
    )
anxiety_scores_df
```

```
## # A tibble: 3 x 3
##   id         text                                                anxiety
##   <chr>      <chr>                                                 <dbl>
## 1 control    yesterday I took my dog for a walk                    0.210
## 2 depression I slept all day and cried in the evening              0.338
## 3 anxiety    I kept thinking of all the things I needed to do      0.354
```

Note that `get_sims()` requires only a dataframe, tibble, or embeddings object with numeric columns; the embeddings can come from any source.

Licensing and Availability

embedplyr is licensed under the GNU General Public License (v3.0). All of its source code is stored publicly on Github (https://github.com/rimonim/embedplyr), with a corresponding issue tracker.

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